

Find more likely buyers within a smaller calling list

ABC SUPERMARKET / GOLD MEMBERSHIP

A classification and customer-prioritization assessment

73.1%

historical responders captured

Top 20%

of customers ranked by score

Soft voting

Selected ensemble

A \$499 offer needs a disciplined calling strategy

\$499

year-end membership offer

Normal price: \$999

20%

discount on purchases

Benefit stated in the assessment

The decision

Which existing customers should ABC call first?

Every call costs money. Calling the whole base is wasteful when only a small minority will ever say yes. The model exists to decide who gets the call.

Predict acceptance



Rank customers



Allocate calls

Target: previous
campaign Response

Follow the evidence from customer data to campaign action

- | | | |
|-----------|---------------------------------|---|
| 01 | Understand the customer data | Data, quality checks, class imbalance, EDA |
| 02 | Build and compare the models | Pipelines, six models, grid search, ensembles |
| 03 | Translate scores into decisions | Errors, ranking, capacity and value |

2,240 customer records connect behavior with acceptance

2,240

customer records

22

source columns

1 or 0

accepted / did not accept

```
df = pd.read_excel("data/marketing_data.xlsx",  
sheet_name="marketing_data")
```

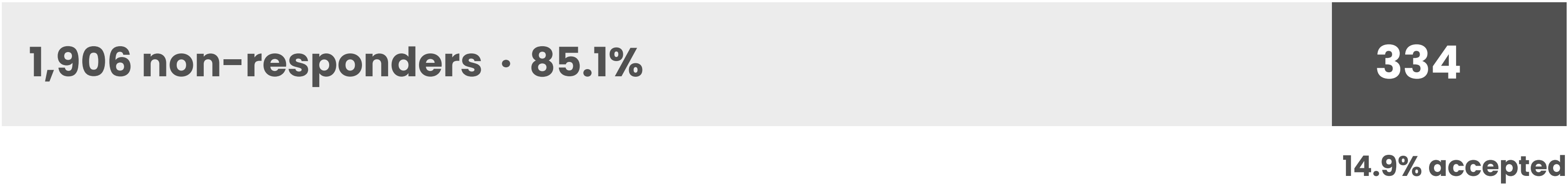
pandas · matplotlib · seaborn

scikit-learn

**Relative path; raw workbook
preserved**

85% accuracy can still identify zero responders

SOURCE TARGET DISTRIBUTION



Dummy accuracy: 85%

Predicting zero for everyone finds no responders.

Primary metric: PR AUC

Notebook “PR AUC” uses average precision, summarizing precision versus recall.

Keep business metrics

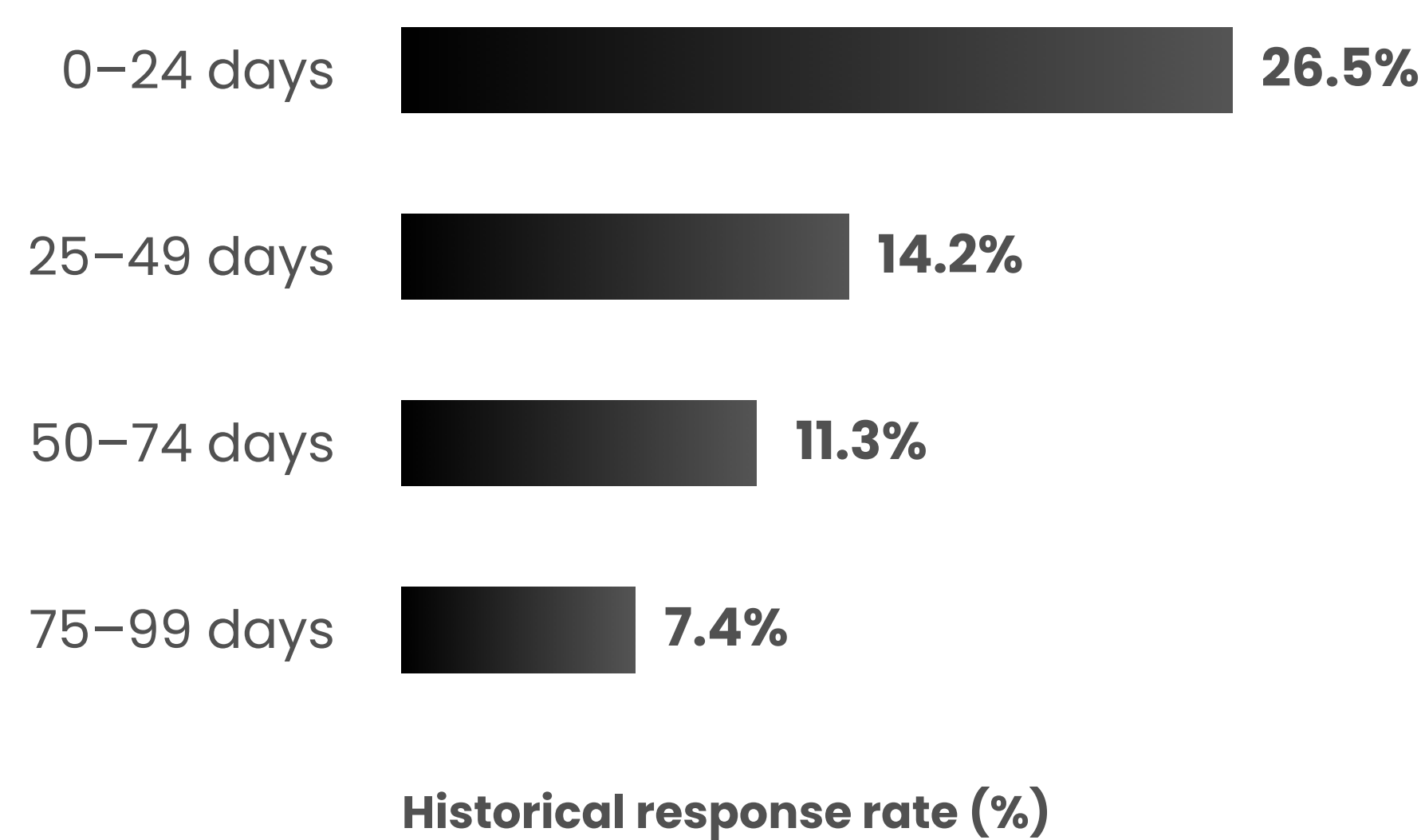
Response at capacity, capture, lift and net contribution.

Four records are excluded; missing income stays for imputation

24	missing incomes	Retain; impute inside the pipeline
3	ages above 100	Exclude using a documented rule
1	income above 200,000	Exclude; sensitivity check still needed
0	duplicate rows / IDs	No deduplication required

2,240 rows → 2,236 retained. Extreme values are flags, not independently verified errors.

Recent purchasers responded more often



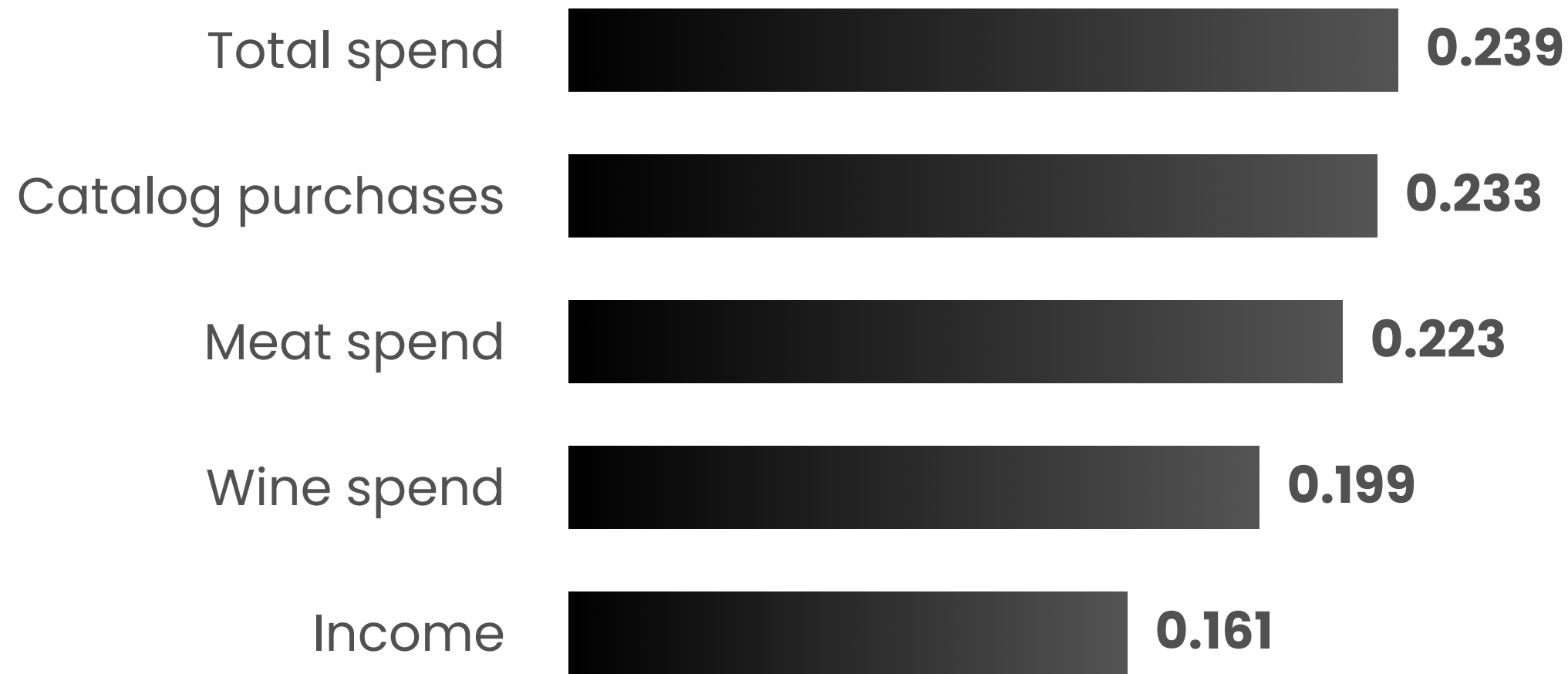
3.6×

response rate in the most-recent versus least-recent quartile

An association worth testing in a model. It does not show that recency causes acceptance.

adobe

Historical spend and catalog activity carry useful signal



Spearman correlation with Response

+0.24

Spearman correlation between total spend and response

These are univariate associations, not final-model feature importance. Total spend is an EDA aggregate only.

Response differs across groups, but sample size matters

Education group	Customers	Response rate
PhD	486	20.8%
Master	370	15.4%
Graduation	1,127	13.5%
2n Cycle	203	10.8%
Basic	54	3.7%

Descriptive differences, not blanket targeting rules. Validate performance across segments.

A stratified split leaves 448 customers for evaluation

2,240

Source rows

2,236

After exclusions

20

Model inputs

1,788 training rows · 80%

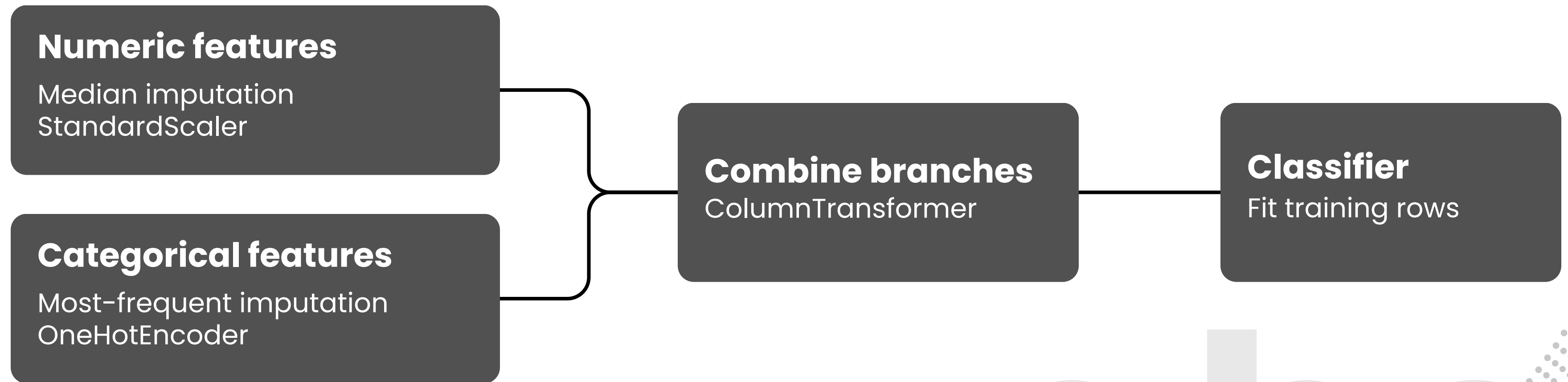
448 rows
evaluation · 20%

Stratified by Response · random_state=42

67 responders in evaluation.

Each validation fold learns preprocessing from its training rows

Inside each outer training fold



Validation rows receive transform + predict only. Scaling mainly benefits logistic regression and KNN.

No initial model wins every campaign tradeoff

Model	Accuracy	Precision	Recall	F1	PR AUC
Dummy	.850	.000	.000	.000	.150
Logistic regression	.812	.427	.746	.543	.624
Decision tree	.717	.319	.791	.455	.437
Random forest	.862	.600	.224	.326	.542
KNN	.859	.600	.179	.276	.376
Gradient boosting	.886	.722	.388	.505	.615

Logistic regression finds more responders; gradient boosting makes more selective predictions.

Five-fold grid search compares 34 candidate configurations

LOGISTIC REGRESSION

6

parameter configurations

$C \times \text{class_weight}$

RANDOM FOREST

12

parameter configurations

$\text{Trees} \times \text{depth} \times \text{min. leaf size}$

GRADIENT BOOSTING

16

parameter configurations

$\text{Trees} \times \text{rate} \times \text{depth} \times \text{min. leaf size}$

34 configurations × 5 folds = 170 CV fits

Training split only

Gradient boosting leads tuned candidates on CV AP

Model	CV AP	Precision	Recall	F1	Eval. PR AUC
Logistic regression	.479	.427	.746	.543	.624
Random forest	.509	.576	.507	.540	.553
Gradient boosting	.538	.500	.806	.617	.589

Tuning does not improve every metric. Boosting recall rises while evaluation AP falls from .615 to .589.

Soft voting averages three complementary response scores

Logistic regression

Linear response patterns

Random forest

Averages bagged trees

Gradient boosting

Sequential residual learning

$$(p_1 + p_2 + p_3) / 3$$

Equal-weight average

Higher score → earlier in the calling list

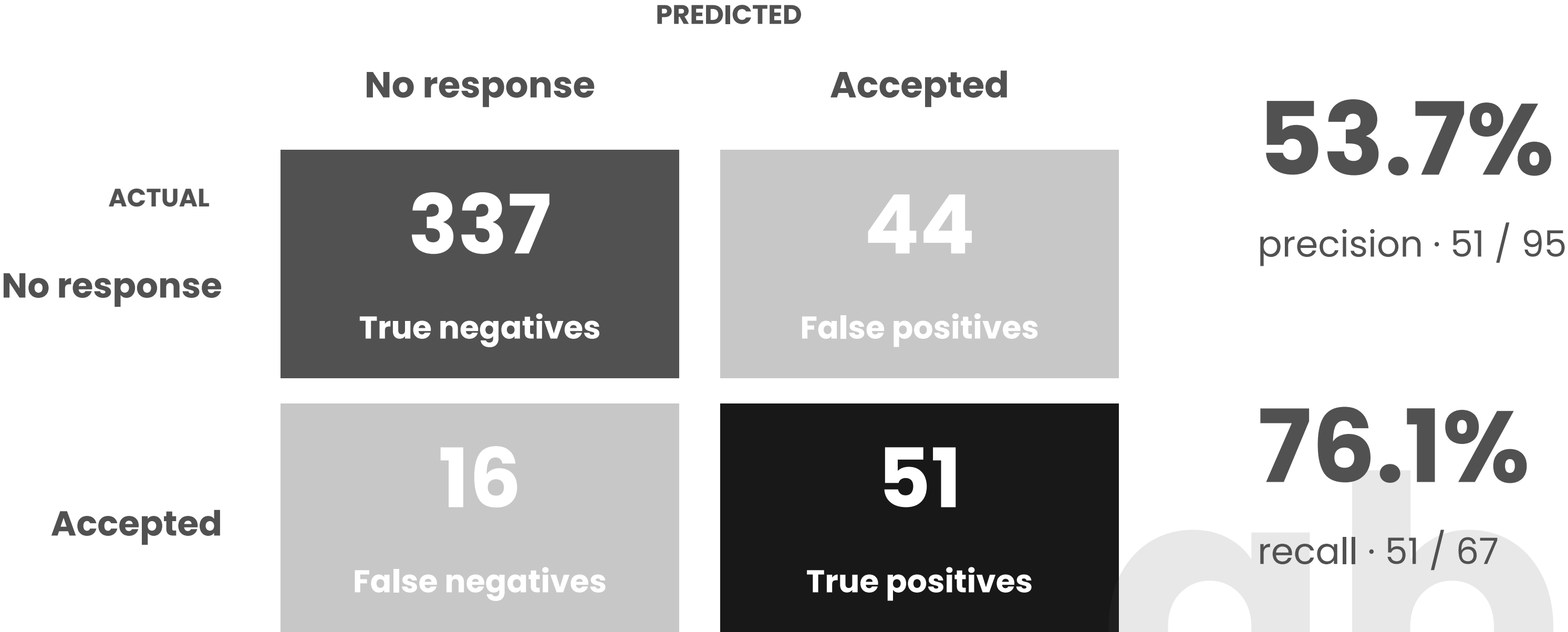
Scores require calibration before they are used as literal purchase probabilities.

Soft voting offers a modest improvement in balance

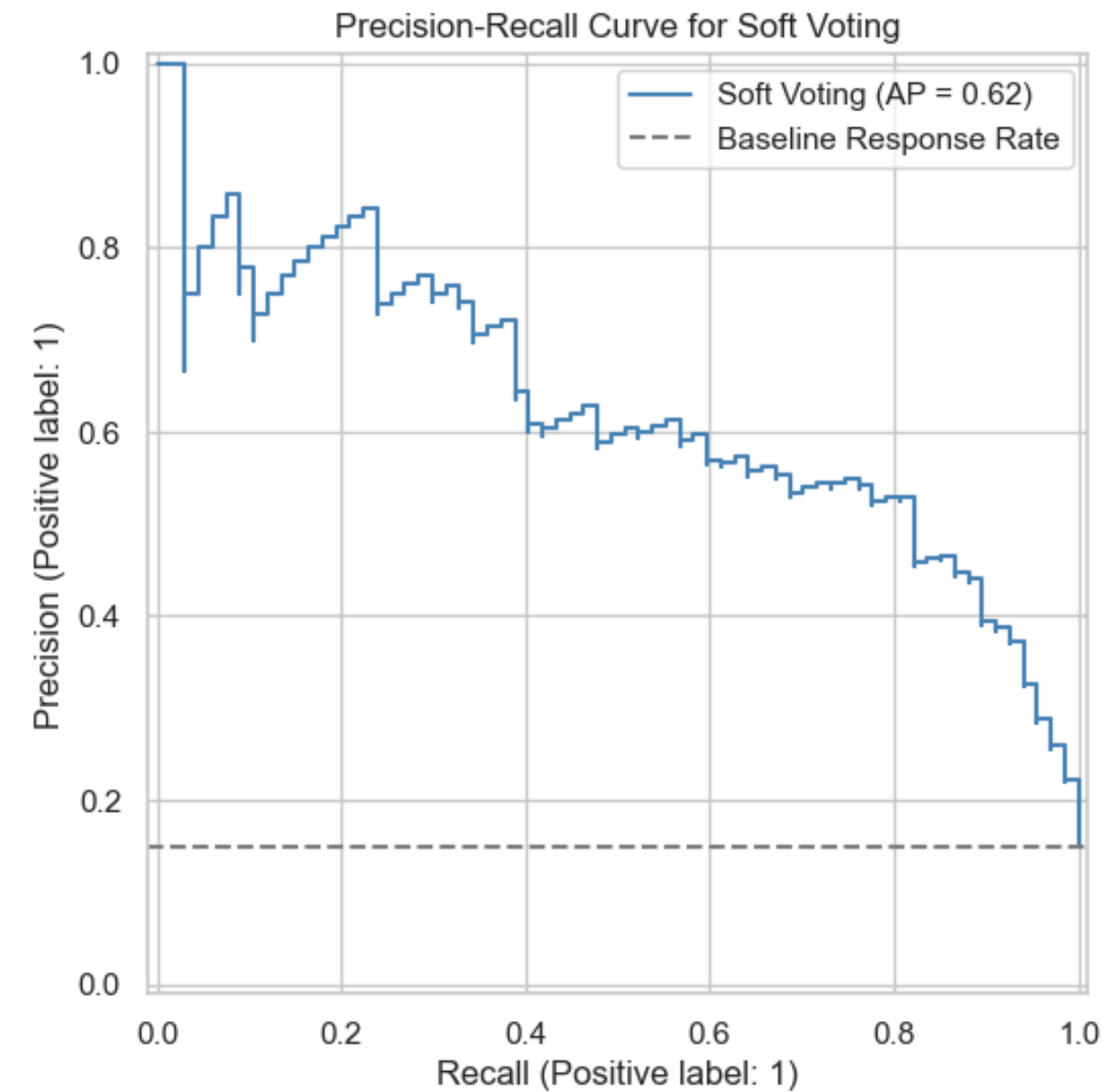
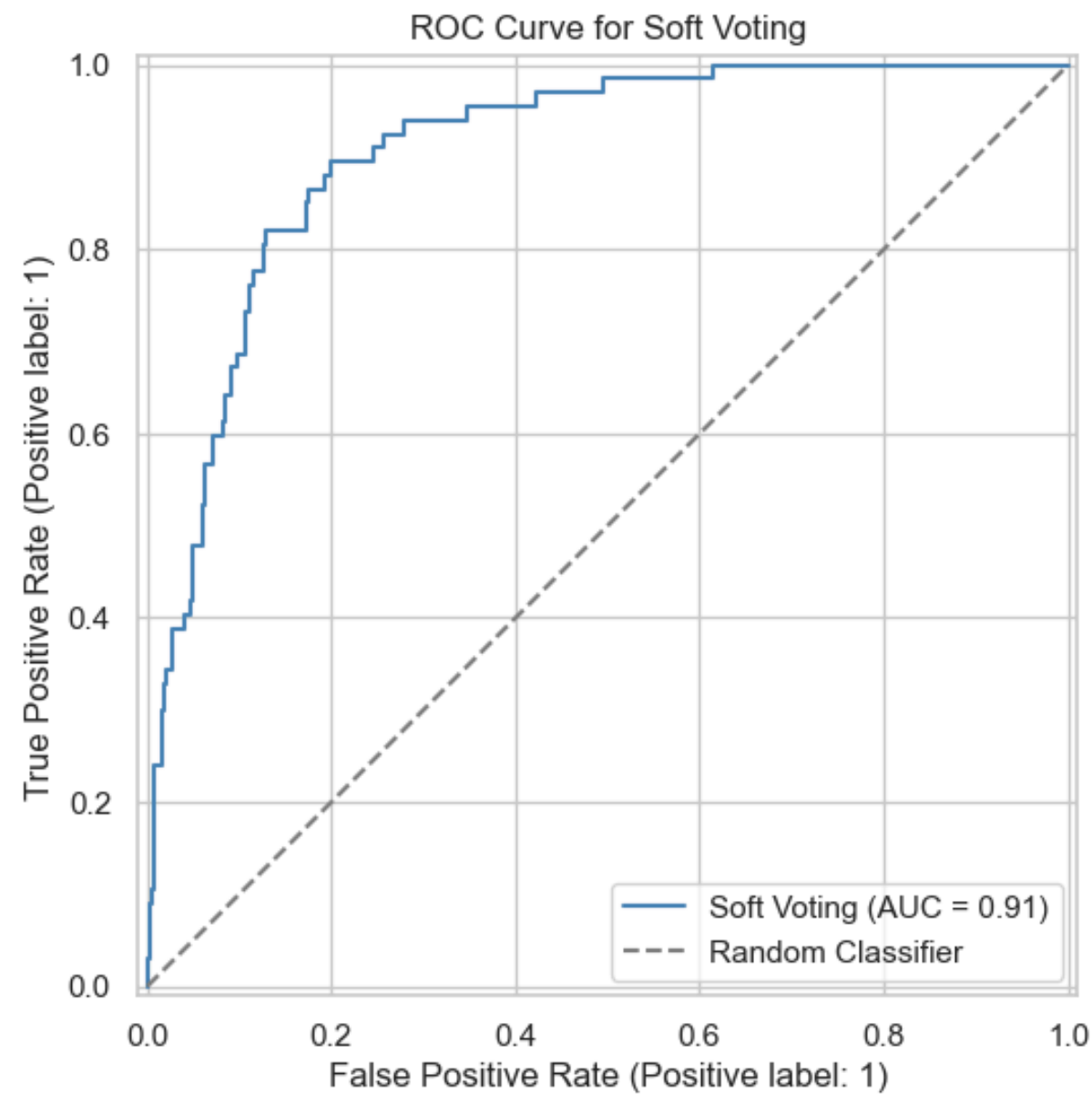
Model	CV AP	Precision	Recall	F1	Eval. PR AUC
Gradient boosting	.538	.500	.806	.617	.589
Extra Trees	.482	.647	.493	.559	.604
Soft voting	.549	.537	.761	.630	.624
Stacking	.545	.457	.866	.598	.616

Soft voting: +.011 CV AP and +.013 evaluation F1 versus boosting; recall declines by 4.5 percentage points.

At 0.50, soft voting finds 51 responders and misses 16



ROC AUC is .907; average precision is .624



Across thresholds; confusion matrix uses one threshold.

Rank by acceptance score, then apply the calling capacity

1. Score

Average the three class-one outputs.

2. Sort descending

Customer A	0.82
Customer B	0.71
Customer C	0.29

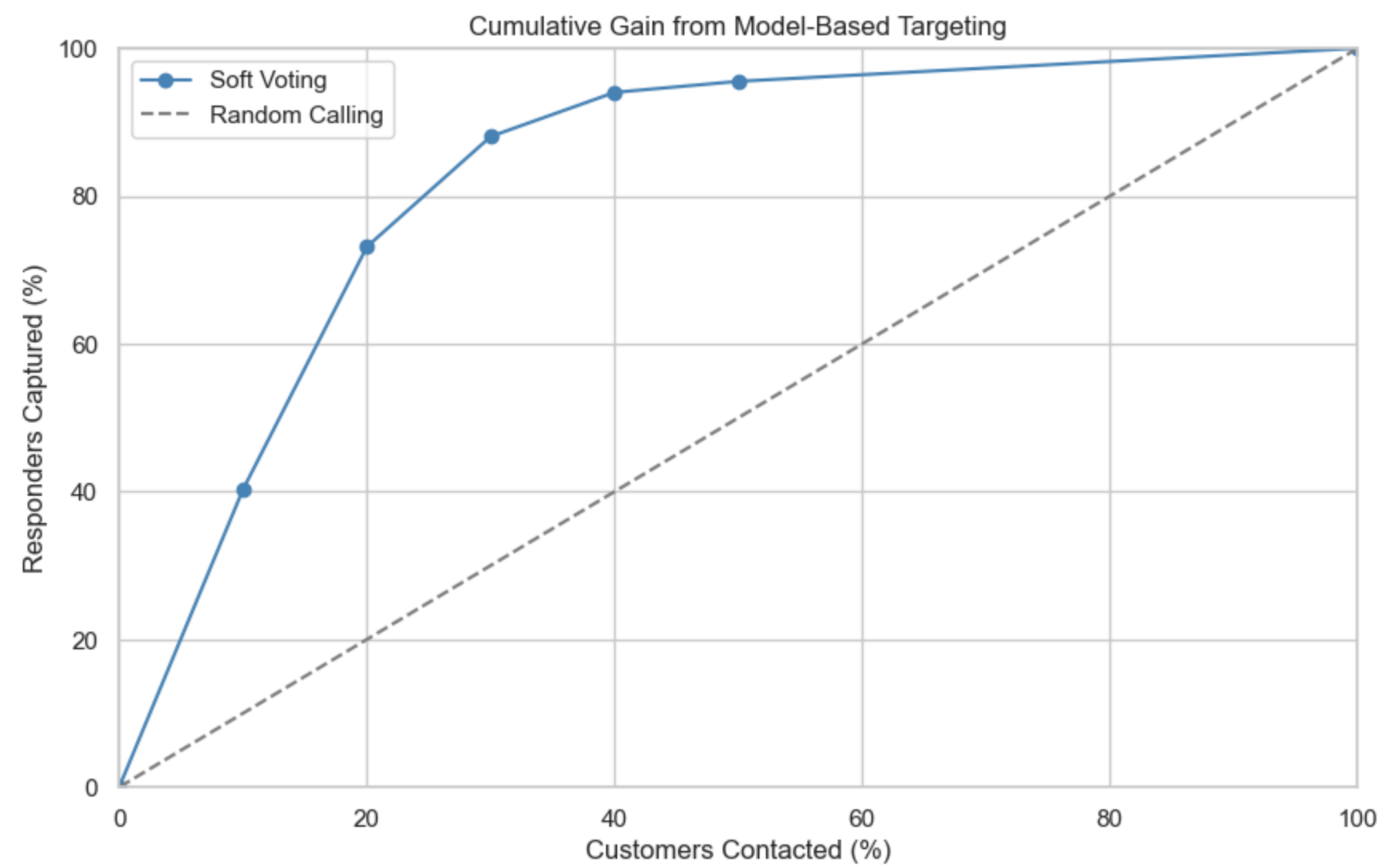
3. Select top k

Call capacity determines the size of the list.

Actual response is used for evaluation only, never for ranking.

Illustrative customer scores above are fictional. Real ranking uses `predict_proba(X_test)[: , 1]`.

The top 20% contains 73.1% of historical responders



49 / 67

responders captured within 90 selected customers

Random benchmark

Calling 20% at random captures 20% of responders in expectation.

A 20% calling budget is a scenario, not a proven optimum

Target share	Calls	Responders	Capture	Response	Lift
10%	45	27	40.30%	60.00%	4.01×
20%	90	49	73.13%	54.44%	3.64×
30%	134	59	88.06%	44.03%	2.94×
40%	179	63	94.03%	35.20%	2.35×
50%	224	64	95.52%	28.57%	1.91×

Moving from 20% to 30% adds 44 calls and 10 historical responders. Choose capacity using cost and value.

Equal call budgets imply a \$17.7k revenue illustration

49

historical responders
in the top 90

13.46

expected in 90
random selections

35.54

difference in
responder labels

$$35.54 \times \$499 \approx \$17,735$$

**Illustrative gross revenue.
No causal or profit claim.**

A randomized pilot should test the targeting policy

Randomly split one eligible customer pool into two policy arms

Model policy

Call the top-scored 20% within its assigned arm.

Random policy

Call a random 20% within its assigned arm.

Same offer, script, budget and observation window

Measure purchases and net contribution per eligible customer. A no-call arm tests contact incrementality.

The next evidence must come from fresh campaign data

- | | | |
|----|--------------------------------|---|
| 01 | Fresh evaluation | Repeated test inspection means reported results are exploratory. |
| 02 | Nested validation | Tune and calculate class weights inside each outer training fold. |
| 03 | Feature timing | Confirm predictors precede the campaign; date cutoff is assumed. |
| 04 | Calibration & drift | Validate scores and economics on a new campaign before scale. |

Pilot soft voting, then decide using conversion and contribution

73.1%

responder capture within the top 20%

The model prioritizes the list. The next campaign validates the value.

01 Freeze the score and contact policy

02 Run a randomized pilot

03 Scale on measured contribution

Which constraint should determine the calling list?

Call capacity

How many customers can the team reach?

Cost of mistakes

What is the cost of an unsuccessful call versus a missed buyer?

Find more likely buyers within a smaller calling list

ABC SUPERMARKET / GOLD MEMBERSHIP

A classification and customer-prioritization assessment

73.1%

historical responders captured

Top 20%

of customers ranked by score

Soft voting

Selected ensemble